EIGHT EQUIVALENT CHARACTERIZATIONS OF REPRESENTABILITY OF THE DIAGONAL BY ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. The three 'formal' pairs are restatements of the same quantified statement at different representation levels; each direction is carried as an explicit obligation with the identity spelled out. The equivalence-invariance input is not carded ('Omitted.') and is declared external.

Allowed standard prerequisites (editorial).

- Isom sheaf as 2-fibre product. Stacks, lemma isom-as-2-fibre-product (cross-chapter, named). Gives (1) iff (2).
- 2-fibre product algebra of categories. Categories, lemmas 2-product-categories-over-C, representable-diagonal-groupoids (cross-chapter, named). The rewriting identities used in (1) iff (3).
- Equivalence invariance of representability. lemma-representable-by-spaces-morphism-equivalent (same chapter, NOT carded, declared external). Gives (3) iff (5) and (4) iff (6).
- Base change by a space-representable category. lemma-base-change-by-space-representable-by-space (carded L26677). Gives (3) implies (7).
- Fibre products of algebraic spaces. Spaces, lemma fibre-product-spaces (cross-chapter, named). Makes the rewritten right-hand side representable.

Source reference. Original reference 02YA: focused-seeds/geometry/S9-alg-representable-diagonal/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0300: sources/stacks-algebraic/pinned-original-source.tex, source lines 900–900. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0457: sources/stacks-algebraic/pinned-original-source.tex, source lines 802–802. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04SI: focused-seeds/geometry/S9-alg-representable-diagonal/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof

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cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $\mathcal{X}$ be a category fibred in groupoids over $(Sch/S)_{fppf}$. The following are equivalent:*

1. *the diagonal $\mathcal{X} \to \mathcal{X} \times \mathcal{X}$ is representable by algebraic spaces,*
2. *for every scheme U over S, and any $x, y \in \mathrm{Ob}(\mathcal{X}_U)$ the sheaf $Isom(x, y)$ is an algebraic space over U,*
3. *for every scheme U over S, and any $x \in \mathrm{Ob}(\mathcal{X}_U)$ the associated 1-morphism $x : (Sch/U)_{fppf} \to \mathcal{X}$ is representable by algebraic spaces,*
4. *for every pair of schemes T_1, T_2 over S, and any $x_i \in \mathrm{Ob}(\mathcal{X}_{T_i})$, $i = 1, 2$ the 2-fibre product $(Sch/T_1)_{fppf} \times_{x_1, \mathcal{X}, x_2} (Sch/T_2)_{fppf}$ is representable by an algebraic space,*
5. *for every representable category fibred in groupoids $\mathcal{U}$ over $(Sch/S)_{fppf}$ every 1-morphism $\mathcal{U} \to \mathcal{X}$ is representable by algebraic spaces,*
6. *for every pair $\mathcal{T}_1, \mathcal{T}_2$ of representable categories fibred in groupoids over $(Sch/S)_{fppf}$ and any 1-morphisms $x_i : \mathcal{T}_i \to \mathcal{X}$, $i = 1, 2$ the 2-fibre product $\mathcal{T}_1 \times_{x_1, \mathcal{X}, x_2} \mathcal{T}_2$ is representable by an algebraic space,*
7. *for every category fibred in groupoids $\mathcal{U}$ over $(Sch/S)_{fppf}$ which is representable by an algebraic space every 1-morphism $\mathcal{U} \to \mathcal{X}$ is representable by algebraic spaces,*
8. *for every pair $\mathcal{T}_1, \mathcal{T}_2$ of categories fibred in groupoids over $(Sch/S)_{fppf}$ which are representable by algebraic spaces, and any 1-morphisms $x_i : \mathcal{T}_i \to \mathcal{X}$ the 2-fibre product $\mathcal{T}_1 \times_{x_1, \mathcal{X}, x_2} \mathcal{T}_2$ is representable by an algebraic space.*

Proof. The equivalence of (1) and (2) follows from Stacks, Lemma 04SI and the definitions. Let us prove the equivalence of (1) and (3). Write $\mathcal{C} = (Sch/S)_{fppf}$ for the base category. We will use some of the observations of the proof of the similar Categories, Lemma 02YA. We will use the symbol $\cong$ to mean "equivalence of categories fibred in groupoids over $\mathcal{C} = (Sch/S)_{fppf}$". Assume (1). Suppose given U and x as in (3). For any scheme V and $y \in \mathrm{Ob}(\mathcal{X}_V)$ we see (compare reference above) that

$$\mathcal{C}/U \times_{x, \mathcal{X}, y} \mathcal{C}/V \cong (\mathcal{C}/U \times_S V) \times_{(x,y), \mathcal{X} \times \mathcal{X}, \Delta} \mathcal{X}$$

which is representable by an algebraic space by assumption. Conversely, assume (3). Consider any scheme U over S and a pair (x, x') of objects of $\mathcal{X}$ over U. We have to show that $\mathcal{X} \times_{\Delta, \mathcal{X} \times \mathcal{X}, (x,x')} U$ is representable by an algebraic space. This is clear because (compare reference above)

$$\mathcal{X} \times_{\Delta, \mathcal{X} \times \mathcal{X}, (x,x')} \mathcal{C}/U \cong (\mathcal{C}/U \times_{x, \mathcal{X}, x'} \mathcal{C}/U) \times_{\mathcal{C}/U \times_S U, \Delta} \mathcal{C}/U$$

and the right hand side is representable by an algebraic space by assumption and the fact that the category of algebraic spaces over S has fibre products and contains U and S.

The equivalences (3) $\Leftrightarrow$ (4), (5) $\Leftrightarrow$ (6), and (7) $\Leftrightarrow$ (8) are formal. The equivalences (3) $\Leftrightarrow$ (5) and (4) $\Leftrightarrow$ (6) follow from Lemma 0457. Assume (3), and let $\mathcal{U} \to \mathcal{X}$ be as in (7). To prove (7) we have to show that for every scheme V and 1-morphism $y : (Sch/V)_{fppf} \to \mathcal{X}$ the 2-fibre product $(Sch/V)_{fppf} \times_{y, \mathcal{X}} \mathcal{U}$ is representable by an algebraic space. Property (3) tells us that y is representable by algebraic spaces hence Lemma 0300 implies what we want. Finally, (7) directly implies (3). $\square$